

03 — Sample Master Design Document (Georgetown-style)

Filled example using the MDD template. Shows the course-header block + one module table. Conventions: $\rightarrow$ CO# = objective laddering; <Created> = artifact to build; curated media lives in the pre-class study guide.

Course-Header Block

Field	Content
Course Name	Introduction to ML Programming
Course Description	For domain experts building proof-of-concept ML apps. Production-ready code over theory; intuitive, visual understanding of algorithms sufficient to make decisions and explain them to stakeholders.
Course-Level Objectives	CO1 (Apply) Implement end-to-end ML pipelines. / CO2 (Evaluate) Select appropriate algorithms. / CO3 (Create) Write production-quality sklearn/Keras code. / CO4 (Analyze) Diagnose failures (overfitting, leakage, imbalance). / CO5 (Apply) Communicate models to stakeholders.
Organization values (mapped)	<i>Quality</i> : every notebook verified runnable. <i>Humanity</i> : welcoming tone, whole-person hints. <i>Integrity</i> : honest about model limits/caveats. <i>Innovation</i> : GenAI-assisted production.
Teaching Principles (applied)	<i>Activate Prior Learning</i> : tie ML to their domain projects. <i>Organize Knowledge</i> : same pipeline every week. <i>Motivate</i> : canonical real datasets. <i>Develop Mastery</i> : gradual-release scaffolding. <i>Practice & Feedback</i> : pair work + post-class with solutions. <i>Climate</i> : welcoming, debugging-is-normal. <i>Self-Directed</i> : cheat sheets, optional deep dives.

Field	Content
Assessment / How Success Is Measured	No grades. Pair-programming + post-class exercises (solutions provided); per-week self-assessment checklists; a capstone project is the ultimate measure.
Materials & Media	Curated videos/readings → each week's pre-class study guide. Frameworks: sklearn (W1–5), Keras 3.x (W6–8).

Module Table

Week 1 — The ML Pipeline & Linear Regression (1 week, 4 hrs)

Module Objectives (→CO#)	Topics (sequential)	Materials & Media	Activities (what students do)
1. (Apply) Execute a full pipeline: load → preprocess → train → evaluate → interpret (→CO1)2. (Apply) Implement linear regression with a proper train/test split (→CO1)3. (Apply) Interpret coefficients in business terms ("holding all else constant") (→CO5)4. (Analyze) Identify residual-plot patterns (scatter, funnel, curve) (→CO4)5. (Evaluate) Judge fit with MSE/RMSE/R ² and communicate it (→CO5)	• The ML pipeline (6 steps; split-before-preprocess) • - Linear regression intuition (best line) • - Loss (MSE) • - Residual plots • - Coefficient interpretation	Curated media → Week 1 pre-class study guide (StatQuest videos; sklearn docs). To build: • <Created> live-session notebook (California Housing) • <Created> pair-programming notebook (fill-in-blank) • <Created> post-class notebook (Diabetes) • <Created> glossary, sklearn cheat sheet	• Pre-class study + warm-up — watch curated videos; attempt warm-up problems. • Live follow-along demo — run the complete California Housing pipeline; read residuals; interpret coefficients. • Pair programming: "Regression Relay" — implement the pipeline on a housing variant (fill-in-the-blank, ~70% provided). • Post-class exercise — linear regression on Diabetes; residual plot + R²; bonus: polynomial features. • Self-assessment — Week 1 "you should be able to..." checklist.

Additional information: Live-session *minute-by-minute timing* is NOT in the MDD — it lives in the Teaching Guide (see `Week1_folder_skeleton/Teaching_Guide/Day1/Segment1_FULLY_SHOWN`). The MDD names *what* the activities are; the teaching guide sequences *how/when* to run them.

Weeks 2–8

...(same header block carries; one module table per week — omitted in this surface-level example)...